

HotStreak Sports Data Scrapping Challenge

HotStreak API used for Scrapping

Sports and Categories API

```
curl --location  
'https://api3.hotstreak.gg/graphql?query=query+system+%7B+system+%7B+__t  
ypename+defaultMargin+maxLegsPerBet+maxMultiplier+pusherCluster+pusherKe  
y+sports+%7B+__typename+id+categories+%7B+__typename+id+createdAt+genera  
tedAt+groupName+name+ordinality+updatedAt+%7D+name+%7D+%7D+%7D&variables  
=%7B%7D&operationName=system' \  
--header 'x-hs3-version: 2'
```

This api will give all the available sports and their corresponding categories.

League Games and Opponents API

```
curl --location  
'https://api3.hotstreak.gg/graphql?query=query+system+%7B+system+%7B+__t  
ypename+defaultMargin+maxLegsPerBet+maxMultiplier+pusherCluster+pusherKe  
y+sports+%7B+__typename+id+categories+%7B+__typename+id+createdAt+genera  
tedAt+groupName+name+ordinality+updatedAt+%7D+name+%7D+%7D+%7D&variables  
=%7B%7D&operationName=system' \  
--header 'x-hs3-version: 2'
```

This api will give a list of all the available games. Each game is provided with its corresponding league information and teams playing that game.

Player and Market API

```
curl --location
'https://api3.hotstreak.gg/graphql?query=query+search%28%24query%3A+String+%24page%3A+Int+%24filters%3A+SearchFilterInput%29+%7B+search%28query%3A+%24query+page%3A+%24page+filters%3A+%24filters%29+%7B+__typename+categoryFilters+dateFilters+gameFilters+leagueFilters+positionFilters+results+%7B+__typename+markets64+participant+%7B+__typename+id+opponentId+player+%7B+__typename+id+createdAt+firstName+fullName+generatedAt+lastName+position+updatedAt+%7D+%7D+%7D+sportFilters+statusFilters+teamFilters+totalCount+%7D+%7D&variables=%7B%22filters%22%3A%7B%22activeMarketsOnly%22%3Atrue%2C%22sport%22%3A%22Z21k0i8vaHMzL1Nwb3J0LzE%22%7D%7D&operationName=search' \
--header 'x-hs3-version: 2'
```

This API will give a list of players and their playing information, and market information. The market information is in protobuf format.

Dynamic Margin For Each Category

Scraping through the JavaScript, I found an Object of arrays containing `historical_inplay_margin` and `historical_pregame_margin` for each category and for each league.

Sample:

```
[{"category_key": "passing_yards", "stat_key": "passYds",
"historical_inplay_margin": -0.0161, "historical_pregame_margin":
0.0068}]
```

I used this data to generate margins as per leagues and categories dynamically.

If the game is live, the margin is `historical_inplay_margin` else `historical_pregame_margin`. Whether the game is live or not is decided by the `market_id` in the Protobuf data.

```
return '1' == market_id.split(':')[2]
```

Market Protobuf Schema

```
// hotstreak_data.proto

syntax = "proto3";

// Defines the data structure for a single probability/odds line within
a market
message Probability {
    double line = 1;           // The player prop line (e.g., 310.5)
    double over = 2;           // Decimal odds for MORE/OVER (e.g., 1.70)
    double under = 3;          // Decimal odds for LESS/UNDER (e.g., 1.60)
    bool suspended = 4;        // Flag indicating if the line is suspended
}

// Defines a single market (e.g., "Passing Yards")
message Market {
    string id = 1;              // Unique ID of the betting line (Odd.id)
    string category = 2;        // Market name (Odd.market)
    repeated Probability probabilities = 3; // Array of odds/line data
    double accrued = 4;         // Likely the current line value (e.g.,
310.5)
}

// Defines the top-level container for all markets in a snapshot
message Markets {
    repeated Market markets = 1; // Array of all Market objects
    string participantId = 2;     // The ID of the match/player the
markets belong to
    int64 snapshotTime = 3;       // Timestamp of the data
}
```

Decimal Odds Calculation

We need to generate decimal odds for both over and under probabilities.

Probability Selection

For each Market, there are multiple Probabilities. Only the median Probability is selected.

```
valid_probabilities.sort(key=lambda p: abs(p.over - p.under))
return valid_probabilities[0]
```

Adjusted Probability by Margins

If the program fails to produce a dynamic margin, a default margin of 0.13 has been set.

$$f = u \times \left(1 + \left(\frac{1 - u}{u} - 0.2 \right) \times e^{-10t} \right)$$

$$\text{Adjusted Probability } (o) = \left(1 + \frac{f}{1 - f} \right) \times t$$

Where,

t = Raw Probability

u = Margin

o = Adjusted Probability

Final Decimal Odds

$$\text{Displayed Decimal Odd } (l) = \frac{1}{\text{Adjusted Probability } (o)}$$

The final decimal Odds are then rounded off to 2 decimal places.